Lab 3
DSA

January 16, 2026

Exercise

You are given an array of n elements.

A modified version of **Merge Sort** is applied to the array using the following rules:

- If the size of the current subarray is greater than or equal to 3, divide it into **three nearly equal subarrays**.
- Recursively apply the same procedure to each subarray.
- When the subarray size becomes less than 3, stop dividing.
- After the recursive calls return, **merge the three sorted subarrays** into a single sorted subarray.

This problem is a modification of Merge Sort in which each recursive call splits the array into **three** parts instead of two.

Time Complexity

Let $T(n)$ denote the running time of the three-way merge sort on an input of size n.

Recurrence. For $n < 3$, the recursion stops immediately:

$$T(1) = \Theta(1), \qquad T(2) = \Theta(1).$$

For $n \geq 3$, the array is divided into three nearly equal parts and merged after recursion:

$$T(n) = T(n_1) + T(n_2) + T(n_3) + \Theta(n),$$

where $n_1 + n_2 + n_3 = n$ and each $n_i \approx n/3$.

For asymptotic analysis:

$$T(n) = 3T\left(\frac{n}{3}\right) + \Theta(n).$$

Master Theorem. The recurrence has the form $T(n) = aT(n/b) + f(n)$ with $a = 3$, since the array is divided into three recursive subproblems, $b = 3$, since each subproblem has size approximately $n/3$, and $f(n) = \Theta(n)$, due to the linear-time merge of the three sorted subarrays. Since

$$n^{\log_b a} = n^{\log_3 3} = n,$$

The recurrence falls under Case 2 of the Master Theorem, yielding

$$T(n) = \Theta(n \log_3 n) = \Theta(n \log n).$$

Note:

Each recursive call corresponds to a node in the recursion tree. Your program must do the following:

- correctly split the array into three subarrays,
- correctly merge the three sorted subarrays to ensure the final array is sorted, and
- correctly print the final sorted array.

Tasks:

(A) Write the logic (set of steps) to solve the given problem.

(B) Write a C program to implement this *three-way merge sort* and determine the final sorted array.[CO4]

Final Output

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Input array:
10 3 7 4 8 2 9 1 6 5

Sorted array:
1 2 3 4 5 6 7 8 9 10
```

Assessment Rubric

Assessment Criteria	**Marks**
Algorithm / Logic (Set of Steps)	2
Correct Three-Way Splitting	3
Correct Three-Way Merging	3
Correct Sorted Output	2
Total	**10**